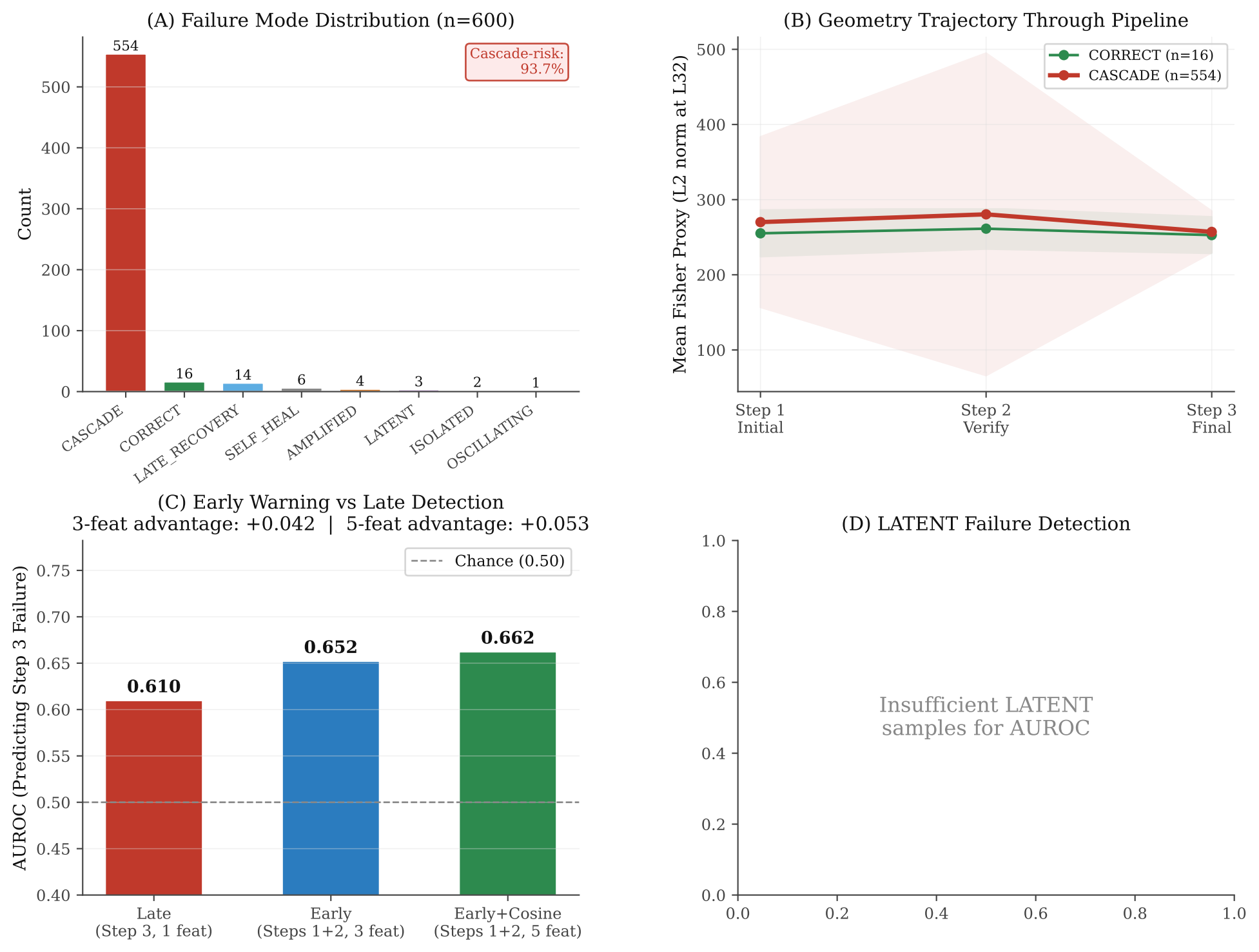
(A) Failure Mode Distribution (n=600)
Count
Cascade-risk: 93.7%
554
16
14
6
4
3
2
1
CASCADE
CORRECT
LATE_RECOVERY
SELF_HEAL
AMPLIFIED
LATENT
ISOLATED
OSCILLATING
(B) Geometry Trajectory Through Pipeline
Mean Fisher Proxy (L2 norm at L32)
CORRECT (n=16)
CASCADE (n=554)
Step 1 Initial
Step 2 Verify
Step 3 Final
(C) Early Warning vs Late Detection
3-feat advantage: +0.042 | 5-feat advantage: +0.053
AUROC (Predicting Step 3 Failure)
Chance (0.50)
0.610
0.652
0.662
Late (Step 3, 1 feat)
Early (Steps 1+2, 3 feat)
Early+Cosine (Steps 1+2, 5 feat)
(D) LATENT Failure Detection
Insufficient LATENT samples for AUROC